Investigate network traffic with Wireshark

Wireshark demo

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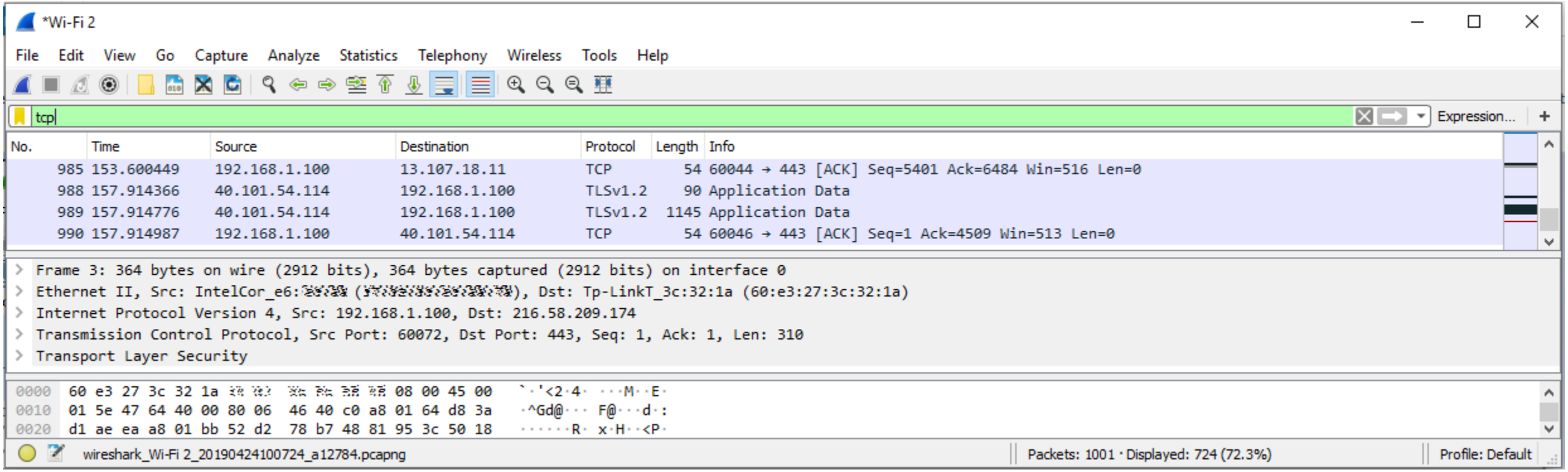
WIRESHARK

Wireshark packet analyzer tool

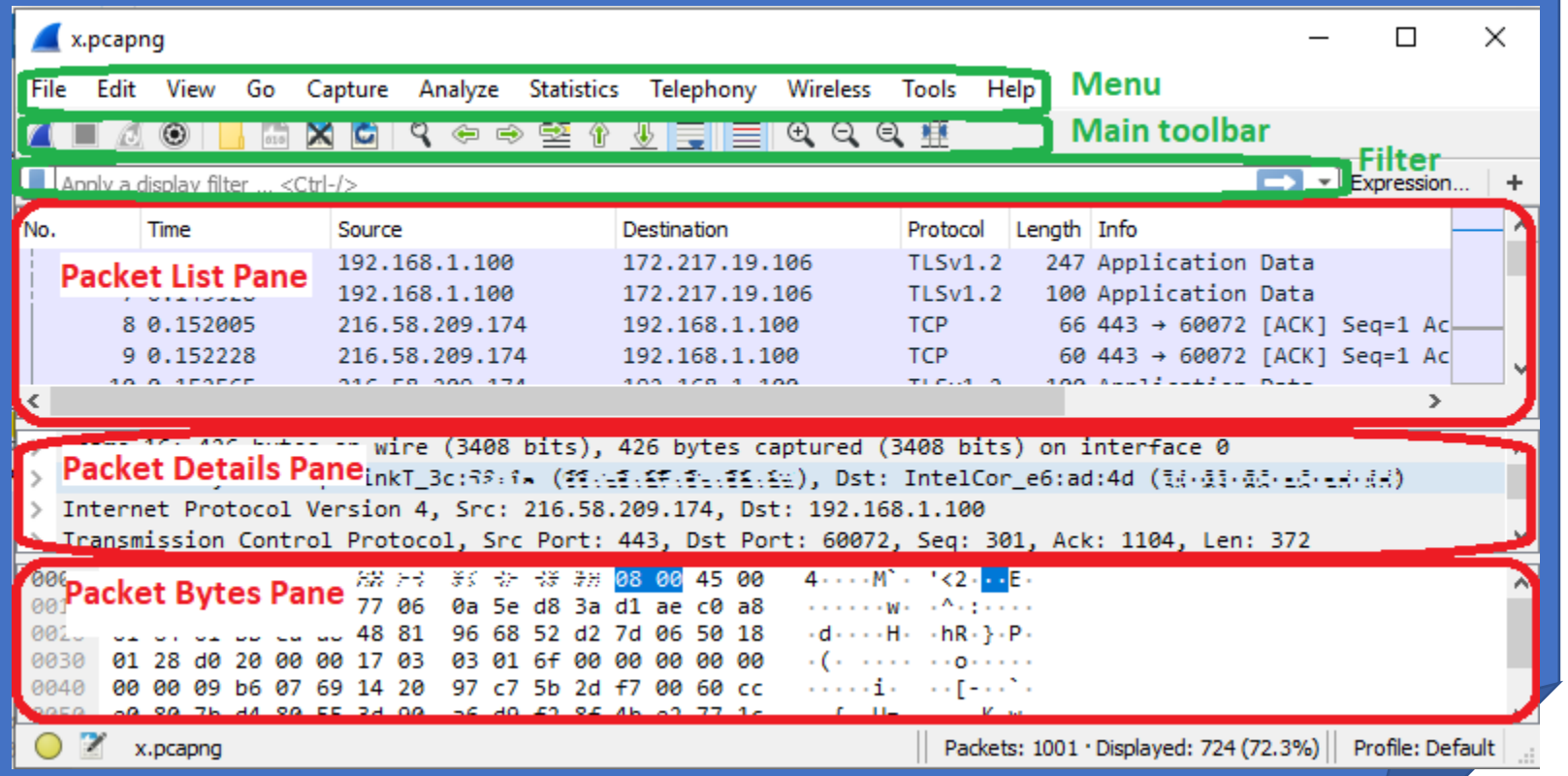
- Analyze logs or live traffic, capture traffic
- Use for network troubleshooting, communication protocol development
- Handle the encapsulation of different protocol layers, display fields, filtering

- Free and open source,
- Easy-to-use, has good documentation



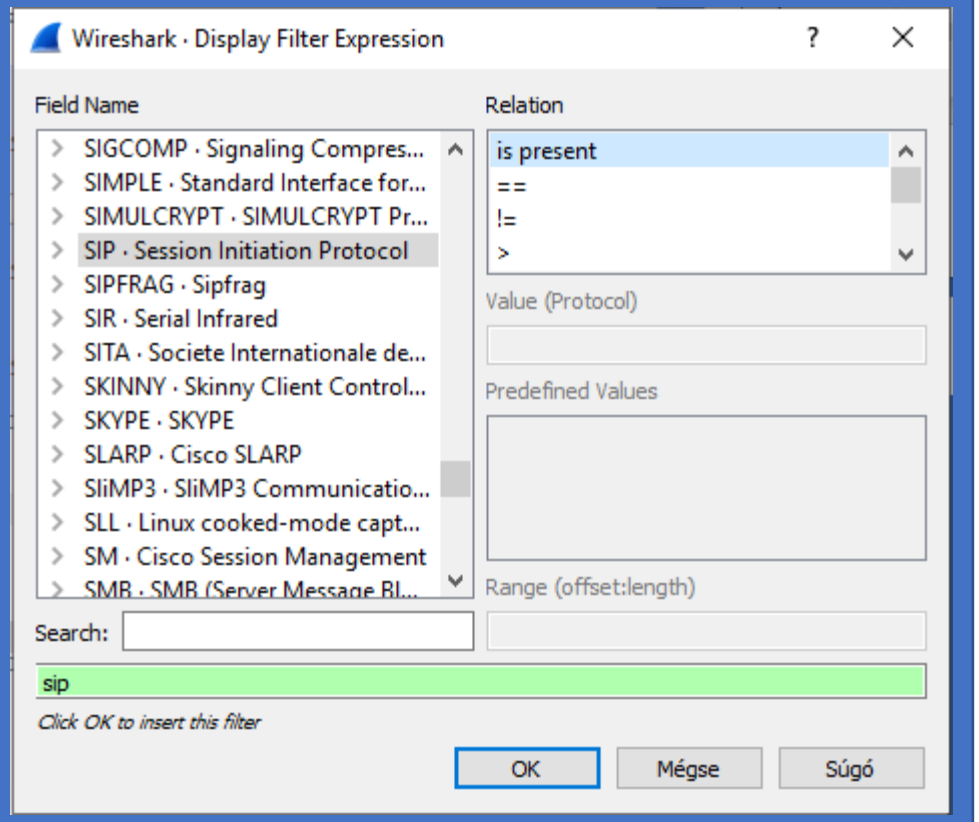
Wireshark demo

1. Download, install and start Wireshark
2. Select an active network connection (WiFi…etc.) *(You can change it later with Capture/Options)*
3. Start / stop capturing network traffic with the and buttons
4. Explore the 3 main views of network traffic:
   - Packet List pane:
     - Identify columns
     - Click on a given column
     - Double click on given row
   - Packet Details pane:
     - Fold/unfold given layers to explore fields
     - Double click on a given element and check the highlight on the Packet Bytes pane
   - Packet Bytes pane:
     - Double click on a given element and check highlight on the Packet Details pane



Wireshark demo

5. Check the Filter toolbar
   - Check preset filters by clicking on Expression...
   - Create your own filters



Examples:

`ip.src==192.168.0.0/16 and ip.dst==192.168.0.0/16`	*Shows only local traffic (source AND destination)*
`ip.addr == 192.168.0.0/16`	*Shows only traffic with local source OR with local destination*
`ip.src != 192.168.1.100`	*Filters out the given source*
`tcp || udp`	*TCP or UDP transport layer protocol*
`tcp.port eq 25`	*Shows only SMTP (=TCP port 25) traffic*
`udp[8:3]==81:60:03`	*Filters UDP packets that contains 0x81, 0x60, 0x03 at the beginning of the UDP payload, skipping the 8-byte UDP header*

Note that you can manage (save/load...etc) filters with the button

Check chapters 6.3. Filtering Packets While Viewing and 6.4. Building Display Fil-ter Expressionsof Wireshark's User's Guide and `wireshark-filter` of Manual Pages to get used to how the filter functionality works. The full list of available filters can be found in the Display Filter Reference